No Extra Epsilon Scaling for Unbounded L^p Graphons

An L^1 sharp-interface theorem and coefficient trichotomy

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Status. Candidate full solution, likely valid; specialist review requested.

1 Source question

Zhang, Scott, Du, and Porter [1] define a graphon Ginzburg–Landau functional on Young measures and prove its sharp-interface limit for bounded graphons. Section 6.2 of arXiv:2408.00422v2 asks how the Dirichlet term should be scaled for unbounded L^p graphons, suggesting that the rate should depend on the singularity of W. Their conclusion again states that this scaling was not determined for general $L^p((0,1)^2)$ kernels. The source takes $p \geq 1$.

2 Idea of the proof

The apparent obstruction disappears once one distinguishes boundedness of the functional from finiteness on its limiting domain. Every L^p graphon in the source regime belongs to L^1. On a binary Young measure, the phase difference is at most 2, so the Dirichlet energy is at most $4\|W\|_1$, even when W is unbounded.

Global continuity of the Dirichlet energy under narrow convergence is neither true nor needed. Truncate the unbounded integrand $|\lambda - \mu|^2$. Narrow convergence of product Young measures gives weak-star convergence for every bounded truncation, which can be tested against $W \in L^1$. Letting the truncation level increase proves lower semicontinuity. The same device handles the quartic double-well potential. The liminf and constant recovery sequence then give the full Gamma-limit. This also classifies every convergent prefactor a_ϵ.

3 Definitions and result

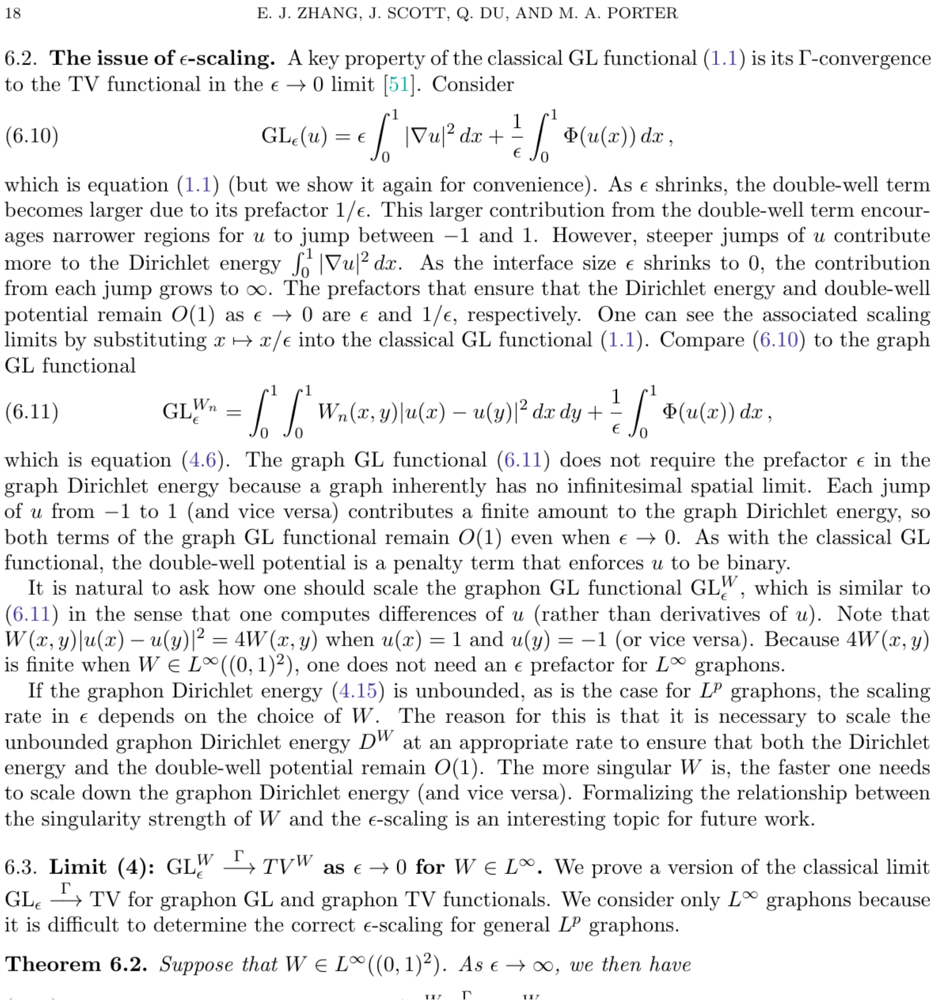
Let $X = (0,1)$ with Lebesgue measure. Let $W : X^2 \to [0,\infty]$ be measurable with $W \in L^1(X^2)$. Write $\mathcal{Y}$ for the source space of parametrized probability Young measures $\nu = \{\nu_x\}_{x\in X}$, equipped with narrow convergence. Define the extended-valued energies

$$D^W(\nu) := \int_{X^2} W(x,y) \int_{\mathbb{R}^2} |\lambda - \mu|^2 \, d\nu_x(\lambda) d\nu_y(\mu) \, dx \, dy,$$

$$V(\nu) := \int_X \int_{\mathbb{R}} \Phi(\lambda) \, d\nu_x(\lambda) \, dx,$$

where $\Phi : \mathbb{R} \to [0,\infty)$ is continuous and

$$\Phi^{-1}(0) = \{-1, 1\}.$$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6.2. **The issue of ϵ-scaling.** A key property of the classical GL functional (1.1) is its Γ-convergence to the TV functional in the $\epsilon \to 0$ limit [51]. Consider

$$\mathrm{GL}_\epsilon(u) = \epsilon \int_0^1 |\nabla u|^2 \, dx + \frac{1}{\epsilon} \int_0^1 \Phi(u(x)) \, dx \,, \tag{6.10}$$

which is equation (1.1) (but we show it again for convenience). As ϵ shrinks, the double-well term becomes larger due to its prefactor $1/\epsilon$. This larger contribution from the double-well term encourages narrower regions for u to jump between -1 and 1. However, steeper jumps of u contribute more to the Dirichlet energy $\int_0^1 |\nabla u|^2 \, dx$. As the interface size ϵ shrinks to 0, the contribution from each jump grows to ∞. The prefactors that ensure that the Dirichlet energy and double-well potential remain $O(1)$ as $\epsilon \to 0$ are ϵ and $1/\epsilon$, respectively. One can see the associated scaling limits by substituting $x \mapsto x/\epsilon$ into the classical GL functional (1.1). Compare (6.10) to the graph GL functional

$$\mathrm{GL}_\epsilon^{W_n} = \int_0^1 \int_0^1 W_n(x,y) |u(x) - u(y)|^2 \, dx \, dy + \frac{1}{\epsilon} \int_0^1 \Phi(u(x)) \, dx \,, \tag{6.11}$$

which is equation (4.6). The graph GL functional (6.11) does not require the prefactor ϵ in the graph Dirichlet energy because a graph inherently has no infinitesimal spatial limit. Each jump of u from -1 to 1 (and vice versa) contributes a finite amount to the graph Dirichlet energy, so both terms of the graph GL functional remain $O(1)$ even when $\epsilon \to 0$. As with the classical GL functional, the double-well potential is a penalty term that enforces u to be binary.

It is natural to ask how one should scale the graphon GL functional GL_ϵ^W, which is similar to (6.11) in the sense that one computes differences of u (rather than derivatives of u). Note that $W(x,y)|u(x) - u(y)|^2 = 4W(x,y)$ when $u(x) = 1$ and $u(y) = -1$ (or vice versa). Because $4W(x,y)$ is finite when $W \in L^\infty((0,1)^2)$, one does not need an ϵ prefactor for L^∞ graphons.

If the graphon Dirichlet energy (4.15) is unbounded, as is the case for L^p graphons, the scaling rate in ϵ depends on the choice of W. The reason for this is that it is necessary to scale the unbounded graphon Dirichlet energy D^W at an appropriate rate to ensure that both the Dirichlet energy and the double-well potential remain $O(1)$. The more singular W is, the faster one needs to scale down the graphon Dirichlet energy (and vice versa). Formalizing the relationship between the singularity strength of W and the ϵ-scaling is an interesting topic for future work.

6.3. **Limit (4): $\mathrm{GL}_\epsilon^W \xrightarrow{\Gamma} TV^W$ as $\epsilon \to 0$ for $W \in L^\infty$.** We prove a version of the classical limit $\mathrm{GL}_\epsilon \xrightarrow{\Gamma} \mathrm{TV}$ for graphon GL and graphon TV functionals. We consider only L^∞ graphons because it is difficult to determine the correct ϵ-scaling for general L^p graphons.

Theorem 6.2. *Suppose that $W \in L^\infty((0,1)^2)$. As $\epsilon \to \infty$, we then have*

Figure 1: The scaling question and the source's L^∞-only theorem on page 18 of arXiv:2408.00422v2.

This includes the source potential $\Phi(t) = (t^2 - 1)^2$. Let

$$\mathcal{Y}^b := \{\nu \in \mathcal{Y} : \nu_x(\{-1, 1\}) = 1 \text{ for a.e. } x\}.$$

For positive coefficients a_ϵ, set

$$F_\epsilon(\nu) := a_\epsilon D^W(\nu) + \frac{1}{\epsilon} V(\nu).$$

Theorem 1 (Integrable-graphon scaling classification)**.** *Let $\epsilon \downarrow 0$.*

(i) *If $a_\epsilon \to a \in [0, \infty)$, then F_ϵ Gamma-converges narrowly to*

$$F_a(\nu) = \begin{cases} aD^W(\nu), & \nu \in \mathcal{Y}^b, \\ +\infty, & \nu \notin \mathcal{Y}^b. \end{cases}$$

(ii) *If $a_\epsilon \to +\infty$, then F_ϵ Gamma-converges narrowly to*

$$F_\infty(\nu) = \begin{cases} 0, & \nu \in \mathcal{Y}^b \text{ and } D^W(\nu) = 0, \\ +\infty, & \textit{otherwise}. \end{cases}$$

In particular, for $a_\epsilon \equiv 1$,

$$\mathrm{GL}^W_\epsilon \xrightarrow{\Gamma} \mathrm{TV}^W$$

for every nonnegative $W \in L^1(X^2)$, where TV^W is exactly the source graphon total variation. Hence the source scaling 1 and $1/\epsilon$ works for every L^p graphon with $p \geq 1$, without dependence on its singularity strength.

4 Proof

We first isolate the only continuity statement that is needed.

Lemma 1. *Both D^W and V are lower semicontinuous for narrow convergence on $\mathcal{Y}$.*

Proof. Suppose $\nu^n \to \nu$ narrowly. For $R > 0$, let

$$q_R(\lambda, \mu) := \min\{|\lambda - \mu|^2, R\}.$$

This is bounded and continuous. Narrow convergence of product Young measures [2, Lemma 8] implies that

$$h_{n,R}(x, y) := \int_{\mathbb{R}^2} q_R(\lambda, \mu)\, d\nu^n_x(\lambda) d\nu^n_y(\mu)$$

converges weak-star in $L^\infty(X^2)$ to the corresponding h_R for ν. Because $W \in L^1(X^2)$,

$$\int_{X^2} W h_{n,R} \longrightarrow \int_{X^2} W h_R.$$

Moreover $D^W(\nu^n) \geq \int W h_{n,R}$. Therefore

$$\liminf_{n\to\infty} D^W(\nu^n) \geq \int_{X^2} W h_R.$$

Letting $R \uparrow \infty$ and using monotone convergence yields

$$\liminf_{n\to\infty} D^W(\nu^n) \geq D^W(\nu).$$

For V, repeat the argument with the bounded continuous truncations $\Phi_R = \min\{\Phi, R\}$, now using the constant test function $1 \in L^1(X)$, and then let $R \uparrow \infty$. □

We prove Theorem 1. Consider arbitrary sequences $\epsilon_n \downarrow 0$ and $\nu^n \to \nu$ narrowly. Observe first that

$$V(\nu) = 0 \quad \Longleftrightarrow \quad \nu \in \mathcal{Y}^b.$$

Indeed, the integrand is nonnegative and its zero set is exactly $\{-1, 1\}$. If $\nu \notin \mathcal{Y}^b$, then $V(\nu) > 0$. Lemma 1 gives $\liminf_n V(\nu^n) \geq V(\nu)$, so

$$\liminf_{n\to\infty} \frac{V(\nu^n)}{\epsilon_n} = +\infty.$$

This proves the Gamma-liminf inequality at every nonbinary ν, for all coefficient regimes.

Suppose now that $\nu \in \mathcal{Y}^b$. If $a_{\epsilon_n} \to a \in (0, \infty)$, then for each $\delta \in (0, a)$, eventually $a_{\epsilon_n} \geq a - \delta$. Lemma 1 gives

$$\liminf_n F_{\epsilon_n}(\nu^n) \geq (a - \delta) D^W(\nu).$$

Let $\delta \downarrow 0$. For $a = 0$, nonnegativity directly gives the required lower bound 0. If $a_{\epsilon_n} \to \infty$ and $D^W(\nu) > 0$, lower semicontinuity implies that $D^W(\nu^n) \geq D^W(\nu)/2$ for all sufficiently large n; hence the liminf is infinite. If $D^W(\nu) = 0$, nonnegativity gives the required lower bound 0.

It remains to construct recovery sequences. If $\nu \in \mathcal{Y}^b$, take the constant sequence $\nu^n = \nu$. Then $V(\nu) = 0$, and

$$D^W(\nu) \leq 4\|W\|_{L^1(X^2)} < \infty.$$

Thus $F_{\epsilon_n}(\nu) = a_{\epsilon_n} D^W(\nu)$, which tends to the stated finite-coefficient limit; it is identically zero in the divergent-coefficient case whenever $D^W(\nu) = 0$. At points where the proposed limit is $+\infty$, the Gamma-limsup requirement is automatic. This proves both parts.

Finally, for $\nu \in \mathcal{Y}^b$, the only possible values of $|\lambda - \mu|$ are 0 and 2, so

$$|\lambda - \mu|^2 = 2|\lambda - \mu|.$$

Consequently $D^W(\nu)$ equals the source definition of $\mathrm{TV}^W(\nu)$, proving the last assertion. □

5 Scope and novelty check

The theorem applies to nonnegative integrable kernels. On the finite-measure square, $L^p \subset L^1$ for every $p \geq 1$, so it resolves the scaling question for the entire L^p-graphon class defined in the source. It does not address signed kernels or genuinely nonintegrable kernels such as $|x - y|^{-1-2s}$; those require a different theory and are not L^p graphons in the stated regime.

The proof deliberately uses lower semicontinuity, not global continuity: moment escape prevents global continuity of D^W on all Young measures. Extended values away from $\mathcal{Y}^b$ are harmless in Gamma convergence.

The bounded novelty search, completed 13 August 2026, covered the run's four lightweight indexes, arXiv:2408.00422v2 and its bibliography, exact-title citation searches, and arXiv-oriented searches combining graphon GL, L^1/L^p, unbounded kernels, Gamma convergence, and epsilon scaling. No later resolution of this extension or coefficient trichotomy was located. The source's 2026 publication still states the problem.

6 Human-review recommendation

Priority specialist review is recommended. The core check is short: verify the product-narrow weak-star convergence for q_R, the L^1 testing step, and the two monotone-convergence passages. A reviewer should also confirm that the intended graphon convention is nonnegative; this is standard and is used throughout the source's energy arguments.

References

[1] E. J. Zhang, J. Scott, Q. Du, and M. A. Porter, *Ginzburg–Landau Functionals in the Large-Graph Limit*, arXiv:2408.00422v2 (2025); Pure and Applied Functional Analysis 11 (2026), 169–209.

[2] A. Braides, P. Cermelli, and S. Dovetta, *Gamma-limit of the cut functional on dense graph sequences*, ESAIM Control Optim. Calc. Var. 26 (2020), article 26.